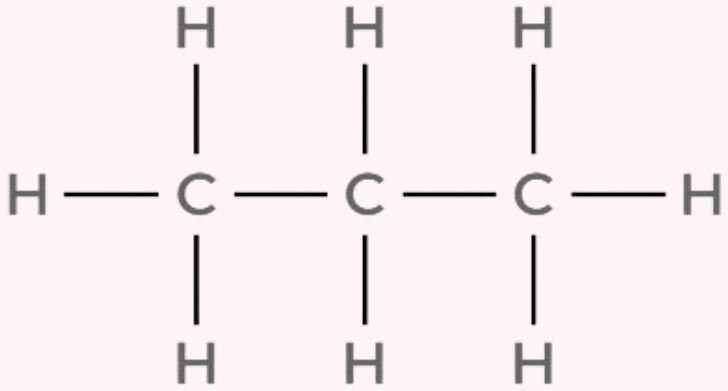
CARD 01 — PROPANE

Molecular Formula: C_3H_8

1. MOLECULAR FORMULA

$$C_3H_8$$

2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$$CH_3CH_2CH_3$$

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

- ☑ Atom count verified
- ☑ Valence checked
- ☑ Bond order checked
- ☑ Same connectivity confirmed

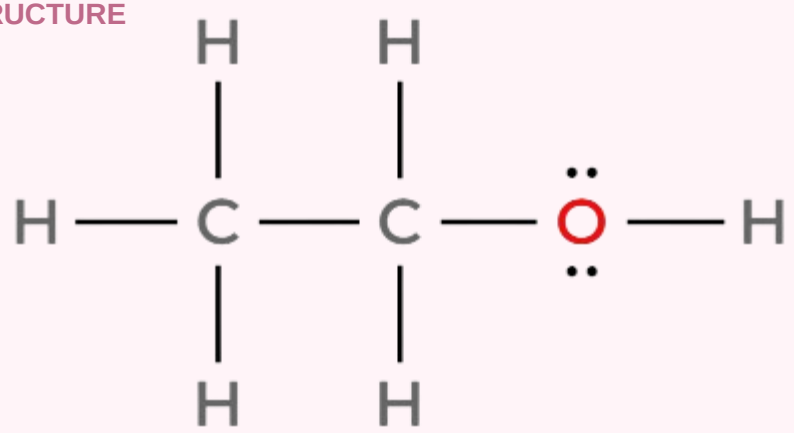
CARD 02 — ETHANOL

Molecular Formula: C_2H_6O

1. MOLECULAR FORMULA

$$C_2H_6O$$

2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$$CH_3CH_2OH$$

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

- ☑ Atom count verified
- ☑ Valence checked
- ☑ Bond order checked
- ☑ Same connectivity confirmed

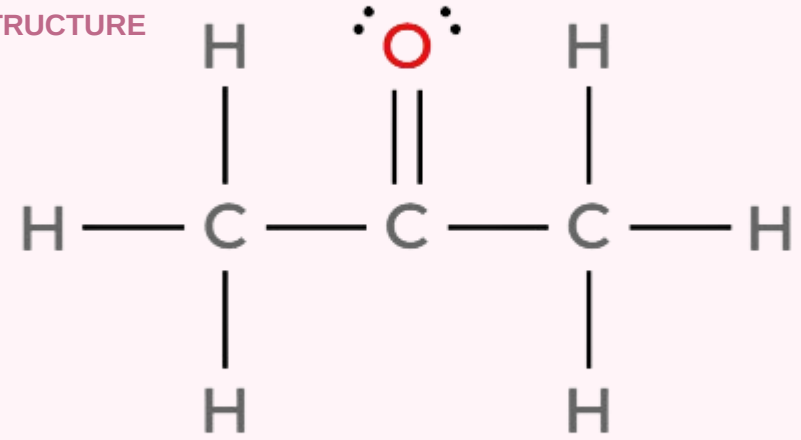
CARD 03 — ACETONE (PROPAN-2-ONE)

Molecular Formula: C_3H_6O

1. MOLECULAR FORMULA

$$C_3H_6O$$

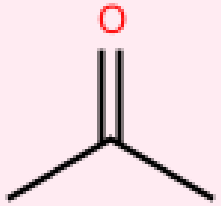
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$$CH_3C(=O)CH_3$$

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

- ☑ Atom count verified
- ☑ Valence checked
- ☑ Bond order checked
- ☑ Same connectivity confirmed

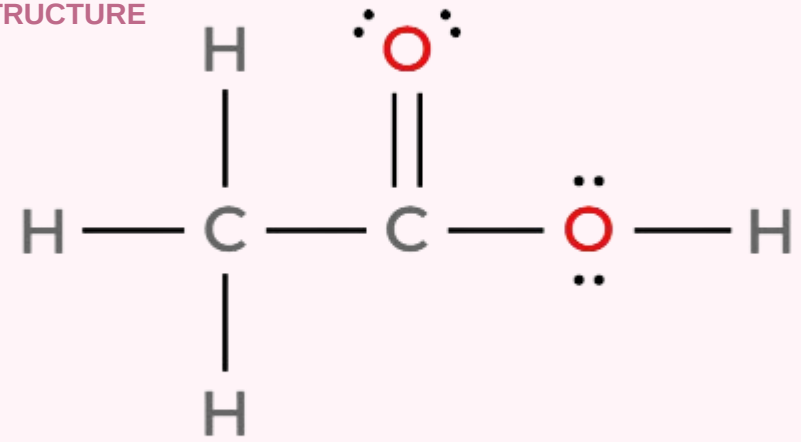
CARD 04 — ACETIC ACID (ETHANOIC ACID)

Molecular Formula: $C_2H_4O_2$

1. MOLECULAR FORMULA

$$C_2H_4O_2$$

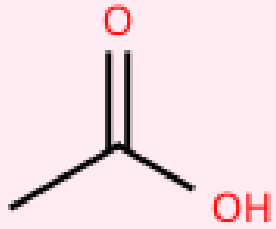
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$$CH_3COOH$$

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

- ☑ Atom count verified
- ☑ Valence checked
- ☑ Bond order checked
- ☑ Same connectivity confirmed

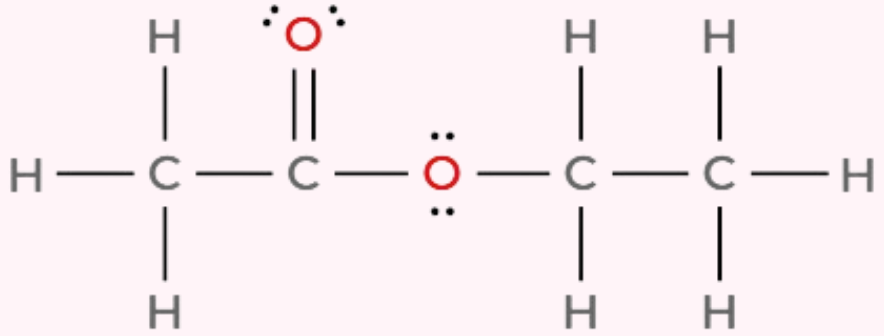
CARD 05 — ETHYL ACETATE (ETHYL ETHANOATE)

Molecular Formula: $C_4H_8O_2$

1. MOLECULAR FORMULA

$$C_4H_8O_2$$

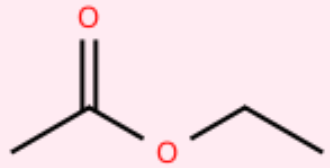
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$$CH_3COOCH_2CH_3$$

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

- ☑ Atom count verified
- ☑ Valence checked
- ☑ Bond order checked
- ☑ Same connectivity confirmed

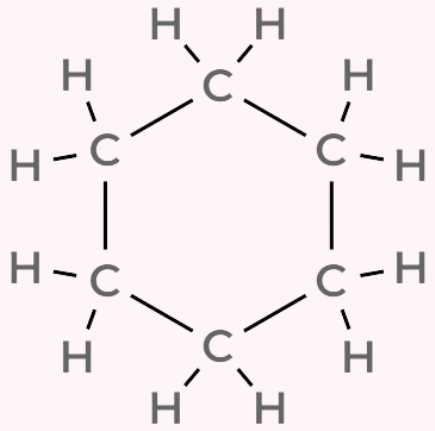
CARD 06 — CYCLOHEXANE

Molecular Formula: C_6H_{12}

1. MOLECULAR FORMULA

$$C_6H_{12}$$

2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$(CH_2)_6$ (ring)

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

- ☑ Atom count verified
- ☑ Valence checked
- ☑ Bond order checked
- ☑ Same connectivity confirmed

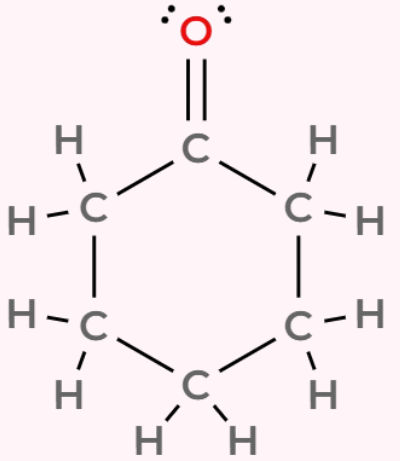
CARD 07 — CYCLOHEXANONE

Molecular Formula: $C_6H_{10}O$

1. MOLECULAR FORMULA

$$C_6H_{10}O$$

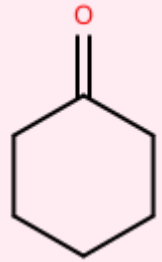
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$$(CH_2)_5C{=}O \text{ (ring)}$$

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

- ☑ Atom count verified
- ☑ Valence checked
- ☑ Bond order checked
- ☑ Same connectivity confirmed

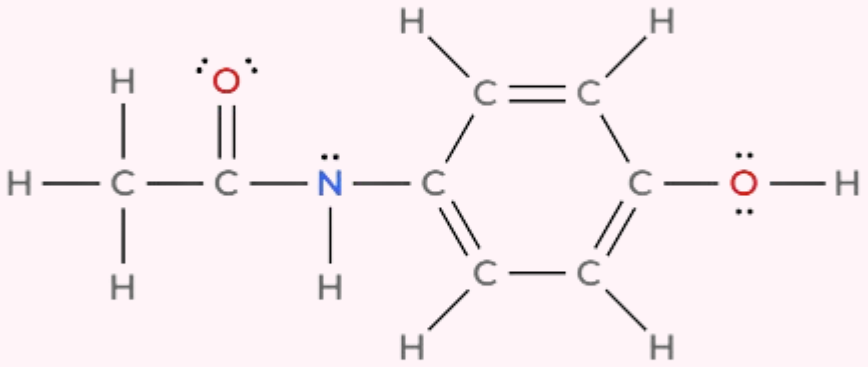
CARD 08 — ACETAMINOPHEN (PARACETAMOL)

Molecular Formula: $C_8H_9NO_2$

1. MOLECULAR FORMULA

$$C_8H_9NO_2$$

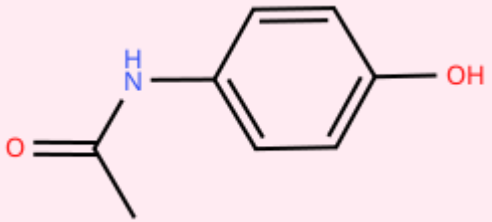
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$$CH_3C(=O)NH–C_6H_4–OH$$

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

- ☑ Atom count verified
- ☑ Valence checked
- ☑ Bond order checked
- ☑ Same connectivity confirmed

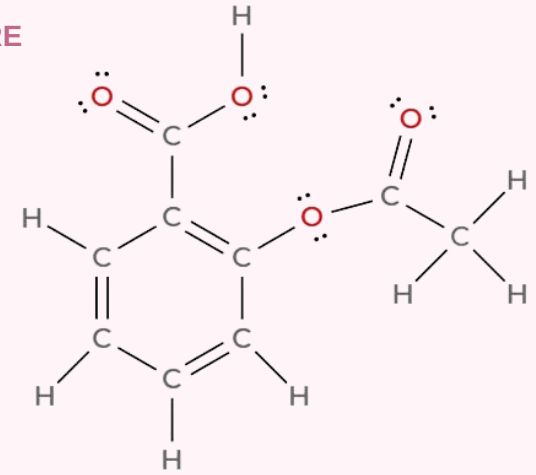
CARD 09 — ASPIRIN (ACETYLSALICYLIC ACID)

Molecular Formula: $C_9H_8O_4$

1. MOLECULAR FORMULA

$$C_9H_8O_4$$

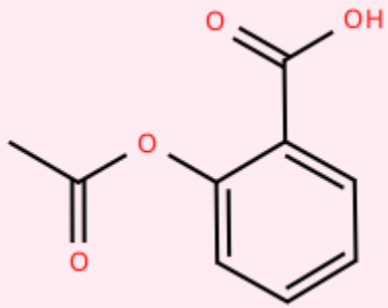
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$$CH_3C(=O)O–C_6H_4–C(=O)OH$$

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

- ☑ Atom count verified
- ☑ Valence checked
- ☑ Bond order checked
- ☑ Same connectivity confirmed

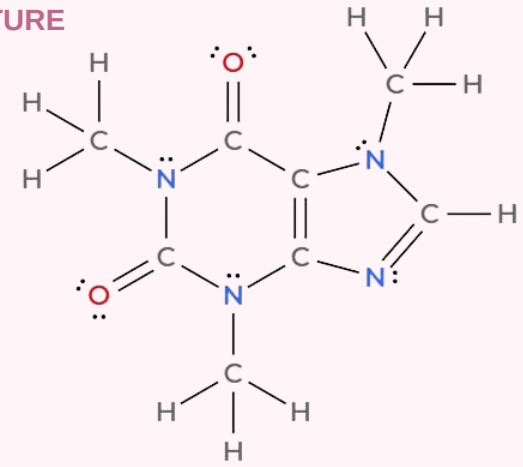
CARD 10 — CAFFEINE

Molecular Formula: $C_8H_{10}N_4O_2$

1. MOLECULAR FORMULA

$$C_8H_{10}N_4O_2$$

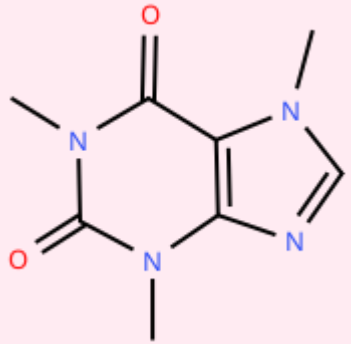
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$$CH_3–N^1–CH=N–C^2=C^1–C(=O)–N(CH_3)–C(=O)–N^2–CH_3$$

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

- ☑ Atom count verified
- ☑ Valence checked
- ☑ Bond order checked
- ☑ Same connectivity confirmed